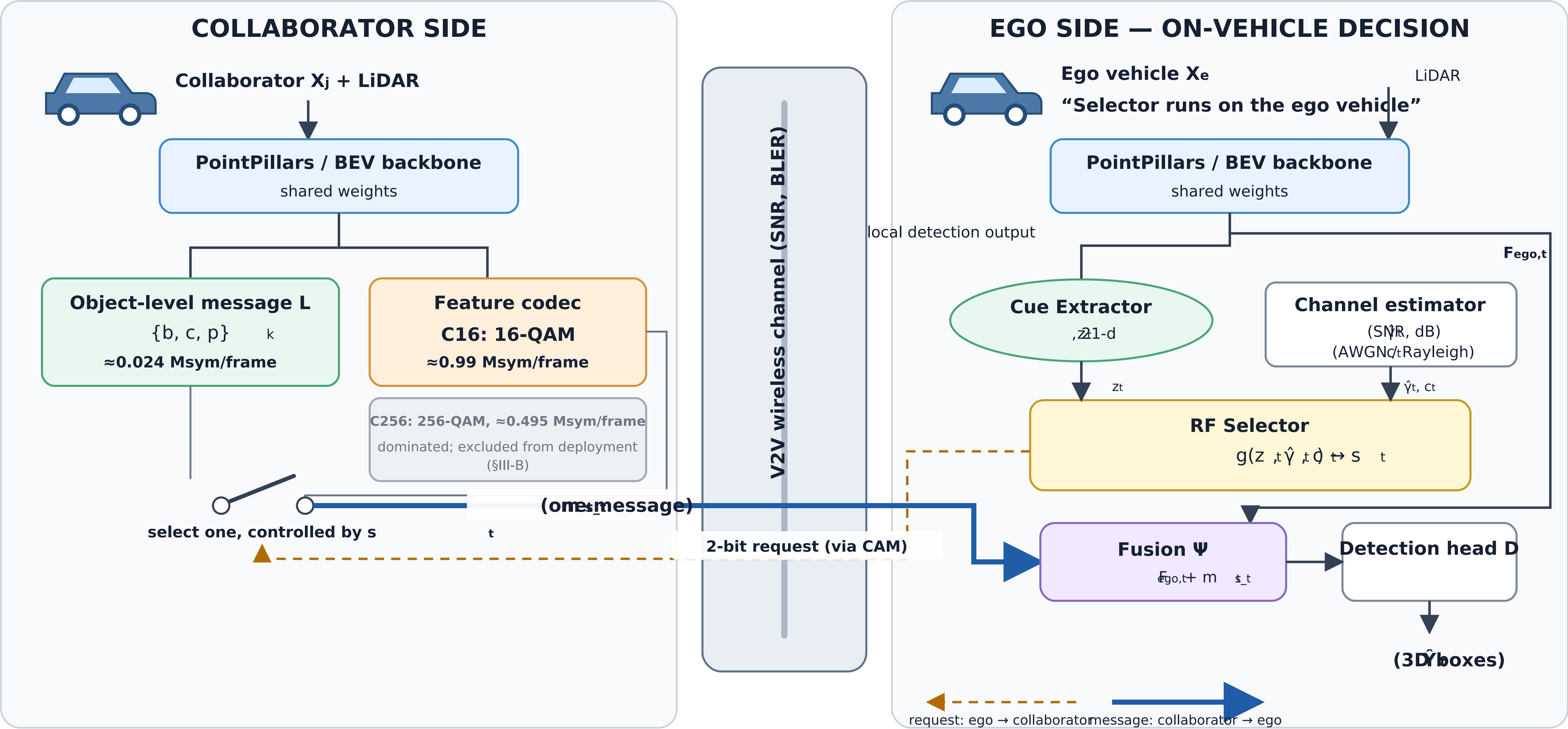
COLLABORATOR SIDE
Collaborator Xj + LiDAR
PointPillars / BEV backbone
shared weights
Object-level message L
{b, c, p}
k
≈0.024 Msym/frame
Feature codec
C16: 16-QAM
≈0.99 Msym/frame
C256: 256-QAM, ≈0.495 Msym/frame
dominated; excluded from deployment
(§III-B)
select one, controlled by s
t
V2V wireless channel (SNR, BLER)
2-bit request (via CAM)
EGO SIDE — ON-VEHICLE DECISION
Ego vehicle Xe
“Selector runs on the ego vehicle”
LiDAR
PointPillars / BEV backbone
shared weights
local detection output
Fego,t
Cue Extractor
Channel estimator
(AWGN / Rayleigh)
zt
γ̂t, ct
RF Selector
Fusion Ψ
Detection head D
request: ego → collaborator
message: collaborator → ego